Exercises: Extreme Values, Quantiles, and Value at Risk

Financial Econometrics

Three exercises covering the extreme-value lectures: RiskMetrics and the risk measures, the block-maxima (GEV) approach and return levels, and clustered extremes. Attempt all three; **full worked solutions are collected at the end**. Throughout, a *loss* is $L = -r$, VaR_p solves $P(L > \mathrm{VaR}_p) = p$, and for the standard normal $z_p = \Phi^{-1}(p)$ with $z_{0.01} = -2.326$, $\phi(z_{0.01}) = 0.0267$.

Exercises

Exercise 1 — RiskMetrics: EWMA volatility, VaR, ES, and diversification

RiskMetrics models the conditional variance by the exponentially weighted moving average

$$\sigma_t^2 = \lambda\,\sigma_{t-1}^2 + (1-\lambda)\,a_{t-1}^2, \qquad \lambda = 0.94,$$

and assumes conditionally normal returns with zero mean over the one-day horizon.

(a) Show that this recursion is the special case of a GARCH(1, 1) model, $\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2$, with $\alpha_0 = 0$, $\alpha_1 = 1 - \lambda$, $\beta_1 = \lambda$. What is the persistence $\alpha_1 + \beta_1$, and why does this make the unconditional variance undefined? What does it imply for the multi-step variance forecast?

(b) Yesterday $\sigma_{t-1} = 1.20\%$ and the return was $a_{t-1} = -3.00\%$. Compute today's volatility σ_t, the one-day 99% VaR, and the one-day 99% Expected Shortfall. (Use $\mathrm{ES}_p = \sigma_t\,\phi(z_p)/p$ for $\mu = 0$.)

(c) A portfolio holds two assets with equal weights $w_1 = w_2 = 0.5$, daily volatilities $\sigma_1 = 1.0\%$ and $\sigma_2 = 1.5\%$, and correlation $\rho = 0.3$. Compute the portfolio 99% VaR from the covariance matrix, and compare it with the weighted sum of the two positions' VaRs. Quantify the diversification benefit.

Exercise 2 — Block maxima, the GEV, and return levels

Let the maximum loss over a block of observations follow the Generalized Extreme Value (GEV) distribution

$$G(x) = \exp\Big[-\big(1 + \xi\,\tfrac{x-\mu}{\sigma}\big)^{-1/\xi}\Big], \qquad \xi \neq 0.$$

(a) The *m-block return level* x_m is the loss exceeded by the block maximum on average once every m blocks, i.e. $G(x_m) = 1 - \frac{1}{m}$. Derive

$$x_m = \mu + \frac{\sigma}{\xi}\Big[\big(-\ln(1-\tfrac{1}{m})\big)^{-\xi} - 1\Big].$$

(b) The shape parameter ξ classifies the tail into three families. Name them for $\xi = 0$, $\xi > 0$, $\xi < 0$, and state which applies to financial losses. For $\xi > 0$, what is the *tail index* and how many finite moments does the loss distribution have?

(c) A GEV is fitted to *monthly* maxima of a stock's daily losses, giving $\hat{\mu} = 2.0\%$, $\hat{\sigma} = 0.8\%$, $\hat{\xi} = 0.25$. Compute the return level for $m = 12$ blocks ($\approx$ one year) and $m = 120$ blocks ($\approx$ ten years). Interpret the ten-year figure in plain words.

Exercise 3 — Clustered extremes and the extremal index

Classical EVT assumes exceedances are independent; real financial extremes *cluster*, and the **extremal index** $\theta \in (0, 1]$ measures the effect. The following 30 days record whether the daily loss exceeded a high threshold (1 = exceedance, 0 = not), in time order:

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0 0 1 1 0 0 0 1 0 0 1 1 1 0 0 0 0 1 0 0 0 0 1 1 0 1 0 0 0 0
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(a) In one or two sentences each, define the extremal index θ and the mean cluster size $1/\theta$. What does Leadbetter's condition $D(u_n)$ ensure, and why do financial returns have $\theta < 1$?

(b) Using *runs declustering* (a cluster is a maximal run of consecutive exceedances; a 0 ends a cluster), count the number of exceedances N_u and the number of distinct clusters. Estimate $\hat{\theta} = (\text{clusters})/N_u$ and the mean cluster size.

(c) The desk uses a one-day 99% VaR as a risk limit. Explain which risk quantities the clustering $\theta < 1$ leaves *unchanged* and which it makes *worse*, and name the standard two-step method that restores independence before applying EVT.

Solutions

Solution 1

(a) Substituting $\alpha_0 = 0,\ \alpha_1 = 1-\lambda,\ \beta_1 = \lambda$ into the GARCH(1,1) variance equation gives $\sigma_t^2 = (1-\lambda)a_{t-1}^2 + \lambda\sigma_{t-1}^2$, exactly the EWMA recursion. The persistence is $\alpha_1+\beta_1 = (1-\lambda)+\lambda = 1$, so the model is **integrated** (IGARCH). The unconditional variance of a GARCH(1,1) is $\alpha_0/(1-\alpha_1-\beta_1)$; here the denominator is 0 (and $\alpha_0 = 0$), so it is **undefined** — shocks never decay. Consequently the multi-step variance forecast does not mean-revert: $\sigma_t^2(h) = \sigma_t^2$ for all horizons h (flat forecast), which is why h-day VaR under RiskMetrics scales exactly like $\sqrt{h}$.

(b) Update the variance (in $\%^2$):

$$\sigma_t^2 = 0.94(1.20)^2 + 0.06(3.00)^2 = 0.94(1.44) + 0.06(9.00) = 1.3536 + 0.54 = 1.8936,$$

so $\sigma_t = \sqrt{1.8936} = \mathbf{1.376}\%$. Then, with $z_{0.01} = -2.326$,

$$\mathrm{VaR}_{0.01} = -z_{0.01}\,\sigma_t = 2.326\times 1.376 = \mathbf{3.20}\%, \qquad \mathrm{ES}_{0.01} = \sigma_t\,\frac{\phi(z_{0.01})}{0.01} = 1.376\times 2.665 = \mathbf{3.67}\%.$$

A single -3% day lifts the volatility from 1.20% to 1.38% — RiskMetrics reacts immediately, the point of an EWMA.

(c) Portfolio variance from $\sigma_p^2 = w^\top \Sigma w$:

$$\sigma_p^2 = (0.5)^2(1.0)^2 + (0.5)^2(1.5)^2 + 2(0.5)(0.5)(0.3)(1.0)(1.5) = 0.25 + 0.5625 + 0.225 = 1.0375,$$

so $\sigma_p = 1.019\%$ and $\mathrm{VaR}_p = 2.326\times 1.019 = \mathbf{2.37}\%$. The two position VaRs are $w_1 z\sigma_1 = 0.5(2.326)(1.0) = 1.163\%$ and $w_2 z\sigma_2 = 0.5(2.326)(1.5) = 1.744\%$, summing to 2.91%. The **diversification benefit** is $2.91 - 2.37 = \mathbf{0.54}$ percentage points: because $\rho < 1$, the portfolio VaR is strictly below the sum of the parts.

Solution 2

(a) Set $G(x_m) = 1 - \frac{1}{m}$ and take logs twice. From $\exp[-(1+\xi\frac{x_m-\mu}{\sigma})^{-1/\xi}] = 1 - \frac{1}{m}$,

$$\big(1+\xi\tfrac{x_m-\mu}{\sigma}\big)^{-1/\xi} = -\ln\big(1-\tfrac{1}{m}\big).$$

Raise both sides to the power $-\xi$: $1+\xi\frac{x_m-\mu}{\sigma} = \big(-\ln(1-\frac{1}{m})\big)^{-\xi}$. Solving for x_m,

$$x_m = \mu + \frac{\sigma}{\xi}\Big[\big(-\ln(1-\tfrac{1}{m})\big)^{-\xi} - 1\Big].$$

(b) $\xi = 0$: **Gumbel** (thin, exponential-like tails); $\xi > 0$: **Fréchet** (heavy, power-law tails); $\xi < 0$: **Weibull** (bounded, finite upper end-point). Financial losses are **Fréchet** ($\xi > 0$). For $\xi > 0$ the tail index is $\alpha = 1/\xi$, and moments $E[L^k]$ are finite only for $k < 1/\xi$ — so the distribution has about $\lfloor 1/\xi \rfloor$ finite moments.

(c) With $\mu = 2.0$, $\sigma = 0.8$, $\xi = 0.25$ (so $\sigma/\xi = 3.2$):

$$m = 12: \quad -\ln(1-\tfrac{1}{12}) = 0.0870, \quad x_{12} = 2.0 + 3.2\big[0.0870^{-0.25} - 1\big] = 2.0 + 3.2(0.841) = \mathbf{4.69}\%,$$

$$m = 120: \quad -\ln(1-\tfrac{1}{120}) = 0.00837, \quad x_{120} = 2.0 + 3.2\big[0.00837^{-0.25} - 1\big] = 2.0 + 3.2(2.307) = \mathbf{9.38}\%.$$

In words: the worst *daily* loss in a typical year is about 4.7%, and the worst daily loss you would expect to see about once a decade is about 9.4%. The ten-year level is far more than double the one-year level — the hallmark of a heavy ($\xi > 0$) tail.

Solution 3

(a) The **extremal index** θ measures the degree of clustering of a stationary series' extremes: $\theta = 1$ means extremes arrive independently (one at a time), while $\theta < 1$ means they come in clusters. Its reciprocal $1/\theta$ is the **mean cluster size** — the average number of consecutive exceedances per episode. Leadbetter's $D(u_n)$ is a mixing condition ensuring that events far apart in time become asymptotically independent at high levels, so the maximum still has a GEV limit (modified to G^θ). Financial returns satisfy $D(u_n)$ but fail the stronger $D'(u_n)$ (no local clustering), because volatility clustering bunches large moves together — hence $\theta < 1$.

(b) Mark the runs of consecutive 1s:

$$00\,\underbrace{11}_{1}\,000\,\underbrace{1}_{2}\,00\,\underbrace{111}_{3}\,0000\,\underbrace{1}_{4}\,0000\,\underbrace{11}_{5}\,0\,\underbrace{1}_{6}\,0000.$$

The number of exceedances is $N_u = 2+1+3+1+2+1 = \mathbf{10}$ and the number of distinct clusters is **6**. Hence

$$\hat{\theta} = \frac{\text{clusters}}{N_u} = \frac{6}{10} = \mathbf{0.60}, \qquad \text{mean cluster size} = \frac{1}{\hat{\theta}} = \frac{N_u}{\text{clusters}} = \frac{10}{6} \approx \mathbf{1.7} \text{ days.}$$

(c) The one-day (marginal) 99% VaR is **unchanged**: it is a quantile of the marginal loss distribution, and θ describes only the *timing* of extremes, not the marginal tail. What clustering makes **worse** are (i) *return levels and multi-period VaR* — the rate of *distinct* extreme episodes is θ times the raw exceedance rate, so the effective sample is $n\theta$, and ignoring θ overstates how often new extremes occur; and (ii) *drawdown risk* — since a breach is typically followed by more (here ~ 1.7 days), the cumulative loss over a cluster is far larger than an independent model implies. The standard fix is **GARCH–EVT (filtered EVT)**: first fit a GARCH to remove the volatility clustering, so the standardised residuals are close to iid ($\theta \approx 1$) and ordinary POT–EVT applies; then rescale the extreme quantile by today's conditional volatility σ_t.